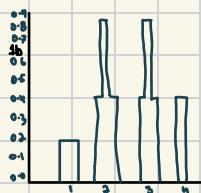
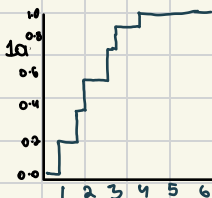


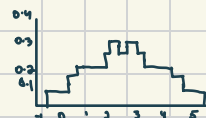
Problem set 4



1	1/6
2	2/6
3	3/6
4	4/6
5	5/6
6	1

$$\frac{1}{2.6 \times 0.2} = 0.1917 \approx 0.19$$

$$1C: 1.34 \cdot 1.1068 \cdot 6^{(-0.2)} = 1.2991$$



1c: when $h=0.2$, the estimate is undersmoothed

with bumps & no density between gaps. when it's

1.299, it's oversmoothed.

$$2a. \hat{f}(0) = \frac{-3}{2 \cdot 0.6} = -\frac{3}{1.2} = -0.5$$

$$2b. E[\hat{f}_h(0)] = \frac{1}{2(0.6)} P(0) \cdot 0.455 = \frac{0.455}{1.2} = 0.3792$$

$$2c. f(0) = 0 \quad \hat{f}_h(0) = \frac{0}{\sqrt{2\pi}} = 0$$

$$2d. E[\hat{f}_h(0)] - f(0) = \frac{S(0) - 1}{\sqrt{2\pi}} = \frac{0.3792 - 1}{\sqrt{2\pi}} = -0.027 \neq 0 \quad X$$

KDE is biased @ 0 & the avg sits below the peak

$$2e. \hat{f}_h(0) = 1 - \frac{0.005 \cdot 0.005}{10} = 0.999$$

$$E[\hat{f}_h(0)] = \frac{0.999}{2.5064} = 0.398$$

$$0.398 - 0.3989 = -0.0009$$

2f: Yes if $h \rightarrow 0$ & h is held fixed, then it stops biased & doesn't converge

2g: small h = low bias & high variance & large is vice versa & is oversmoothed

$$3a. \hat{f}_h(x) = \frac{\sum_{i=1}^n \mathbb{I}\{x_i - x \in [-h, h]\}}{2(5)(15)}$$

3c: The biggest jump is 80 to 85

with a diff of 0.133. It exists bc

the kernel is box & the point of the neighbor

As h & S increases the Gaussian kernel is

$$3b. \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \hat{f}_h(x) = \frac{1}{\sqrt{2\pi}} \sum_{i=1}^S \phi\left(\frac{x_i - x}{1.5}\right)$$

changes smoothly

$$\hat{f}_h(x) = \frac{1}{\sqrt{2\pi}} \left[\text{hist} \right] = 0.152$$

$$(45) = 0.1687$$

$$70 = 0.1790$$

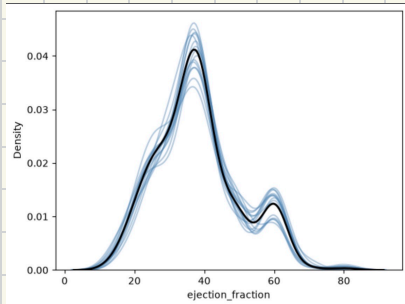
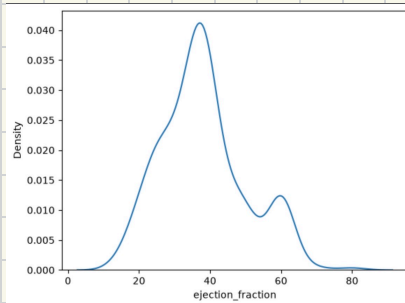
$$80 = 0.1674$$

$$85 = 0.1506$$

3d: Uniform is easy to compute & Gaussian helps

give smooth results so I would use Gaussian for smooth

estimation & uniform when we want a simple window.



4c: The original is reliable between 0-20 & 45-55. It's least reliable at the peaks

especially the biggest peak at 37-38 & at 60, and also at 25